4.6 Additional Deliverables

Platform: **EduHub** (Online Learning Platform) | Track: Full Stack .NET Web Development

1. RESTful API Endpoint Documentation

Method	Route URI	Payload / Parameters	Description & Access Scope
GET	`/api/courses`	`?search=&category=&page=1&size=10`	Retrieves paginated course catalog. (Public)
GET	`/api/courses/{id}`	Route param: `id (int)`	Retrieves syllabus & instructor profile. (Public)
POST	`/api/lessons/{id}/media-token`	Route param: `id (int)`	Issues temporary SAS streaming token. (Student)
POST	`/api/lessons/{id}/progress`	Body: `{"isCompleted": true}`	Updates lesson state & recalculates course progress.
POST	`/api/quizzes/{id}/submit`	Body: `{"answers": [{"qId": 1, "aId": 2}]}`	Evaluates score against passing threshold. (Student)
POST	`/api/checkout/create-session`	Body: `{"courseIds": [1], "coupon": "PROMO"}`	Calculates totals & returns Stripe session URL.
POST	`/api/webhooks/stripe`	Payload from Stripe with signature header	Atomically creates order & enrollment records.
GET	`/api/certificates/verify/{guid}`	Route param: `guid (UUID)`	Returns cryptographic verification of certificate. (Public)

2. Quality Assurance & Testing Strategy

- **Unit Testing (xUnit & Moq):** Covers isolated domain business rules in `EduHub.Services`. Validates discount strategy implementations (`PercentageDiscountStrategy`, `FixedAmountDiscountStrategy`), certificate hash generation, and progress boundary clamps.
- **Integration Testing:** Built with `Microsoft.AspNetCore.Mvc.Testing` using EF Core InMemory and local SQL instances to ensure atomic commits across multi-repository Unit of Work transactions.
- **User Acceptance Testing (UAT):** Validates full customer journey: registration → course discovery → Stripe checkout → SAS video playback → quiz assessment → dynamic PDF certificate download.

3. Cloud Deployment Strategy & CI/CD Pipeline

- **Hosting Infrastructure:**
  - **App Server:** Azure App Service running Linux (.NET 8 LTS environment) configured with auto-scale triggers based on CPU utilization (>70%).
  - **Database Engine:** Azure SQL Database provisioned with Transparent Data Encryption (TDE), automated daily backups, and indexed foreign keys.
  - **Asset Storage:** Azure Blob Storage utilizing Hot access for thumbnails and private containers with SAS tokens for video assets.
- **CI/CD Workflow (GitHub Actions):**
  - **Automated Pipeline:** Triggered on commits to `develop` and pull requests to `main`. Executes build restore, code quality analysis, and xUnit test suites.
  - **Zero-Downtime Deployment:** Publishes compiled artifacts to an Azure *Staging Deployment Slot*, runs automated health checks, and swaps to *Production* seamlessly with zero user interruption.